Microeconomics
Professor Sara Negrelli
L2

Supply & Demand
Competitive markets

Competitive markets

- Market Demand
- Market Supply
- Market Equilibrium:
  - Equilibrium price and quantity
  - How shocks to demand and/or supply alter the equilibrium price and quantity
- Demand and Supply Elasticity

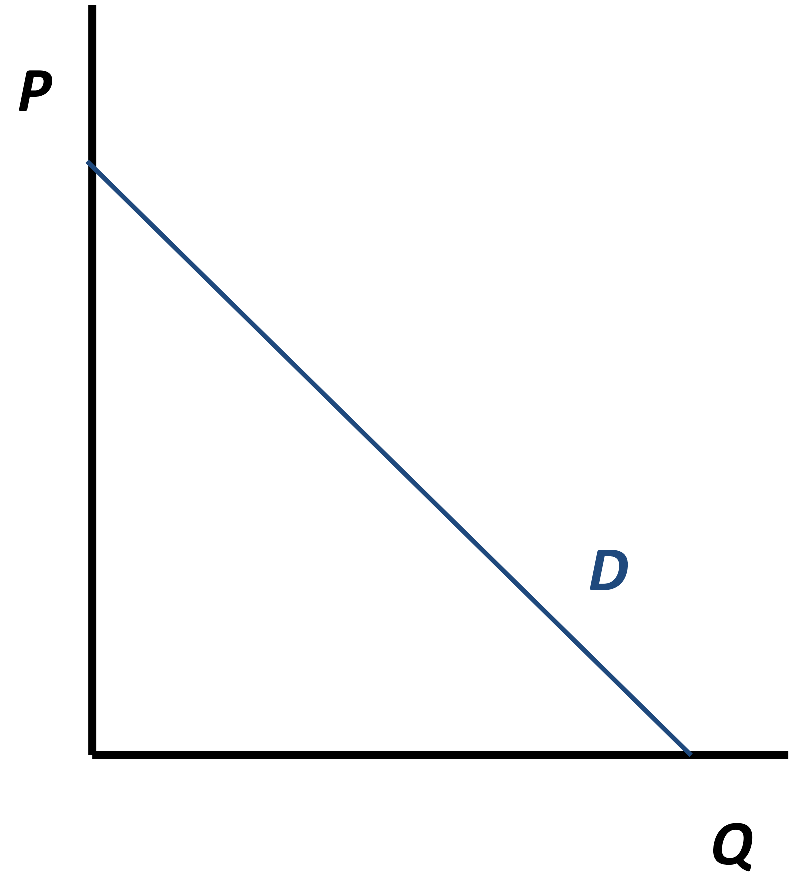
Determinants of Demand

- Demand for a good depends on:
  - Price of the good
  - Prices of other related products
    - Substitutes (an increase in the price of one product causes buyers to demand more of the other, all else equal)
    - Complements (an increase in the price of product causes buyers to demand less of the other, all else equal)
  - Income
    - Normal good: if income increases, quantity demanded increases
    - Inferior good: if income increases, quantity demanded decreases
  - Consumer tastes and preferences
  - Population growth (demographics)
  - Government taxes or regulations
  - Other determinants…

Market Demand Curve

- Market demand curve of a good shows:

  how much of the good consumers want to buy at each possible price *holding fixed all other factors that affect the demand*
- Downward sloping (buying the product is less attractive when the price is high than when the price is low)
- Where does it come from? Individual demand curves (individual preferences, utility maximization)



Shifts and Movement along the Demand Curve

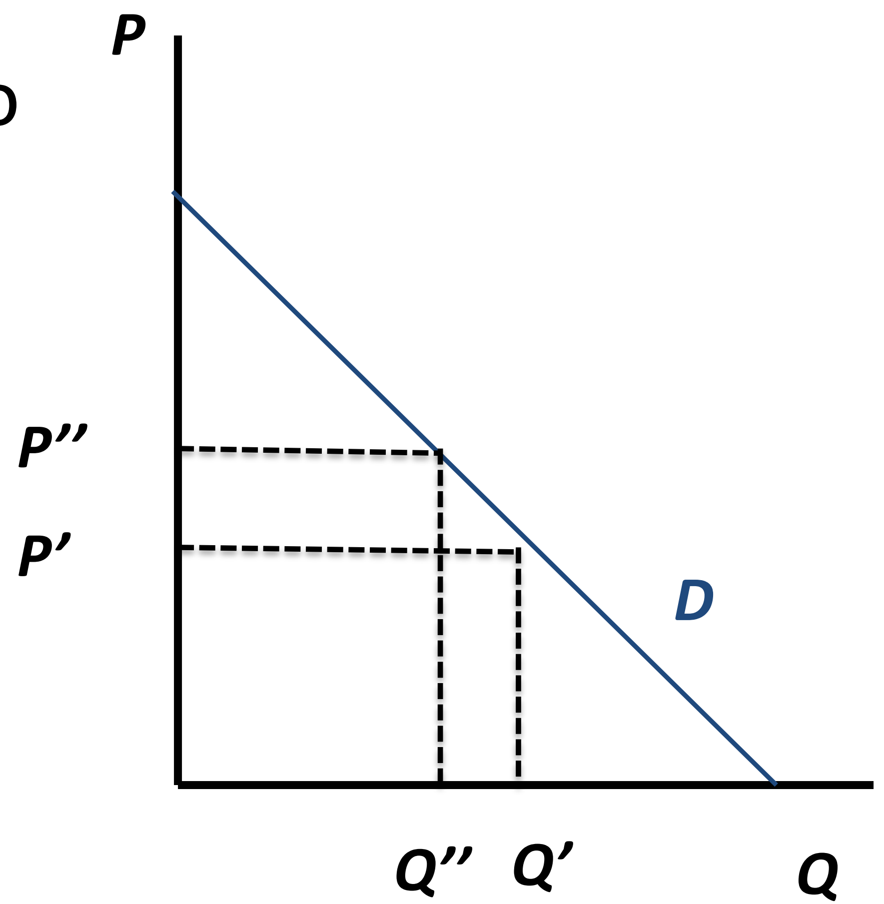
- Change in price of the product causes a *movement along* the demand curve
  - A change in the quantity demanded

- Change in another factor causes the entire demand curve to *shift*
  - A change in the quantity demanded at every price

Movement along the Demand Curve

Increase in price from p’ to p’’

➔ Q decreases from q’ to q’’



Demand Shifts

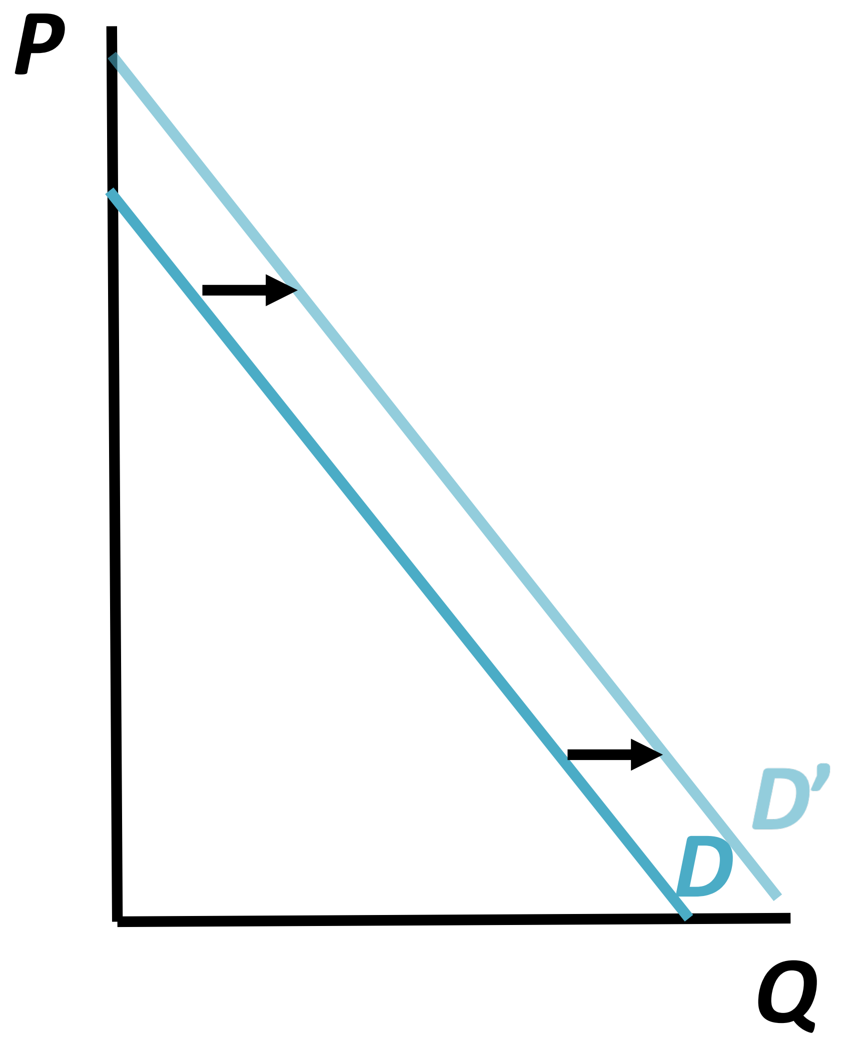
What happens to the demand for Samsung phones when the price of Apple Iphone increases?

Demand Shifts

Substitute products:

$\uparrow P_{iphone} \Rightarrow \uparrow D_{samsung}$

→*Demand for Samsung shifts right (increases)*



Demand Shifts

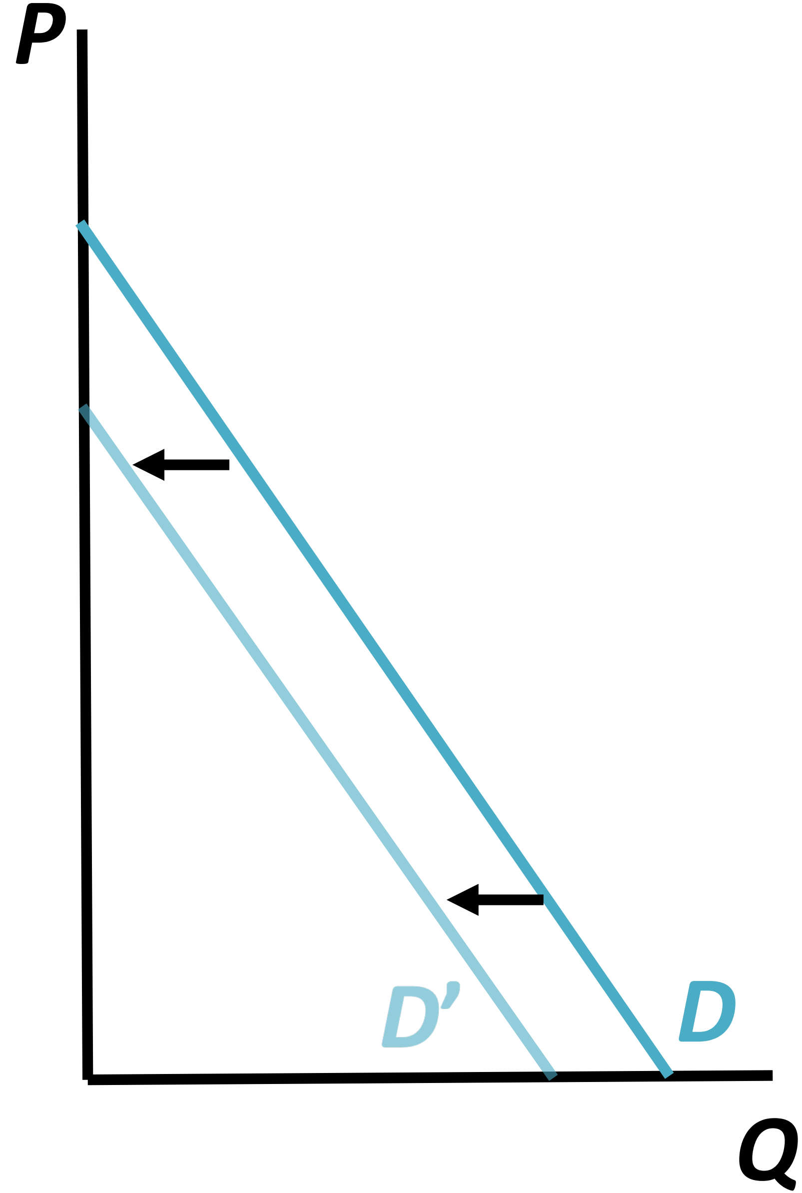
What happens to the demand for polaroids when the price of polaroid films increases?

Demand Shifts

Complement products:

$\uparrow P_{polaroidfilms} \Rightarrow \downarrow D_{polaroid}$

→*Demand for Polaroid shifts left (decreases)*



Demand Shifts

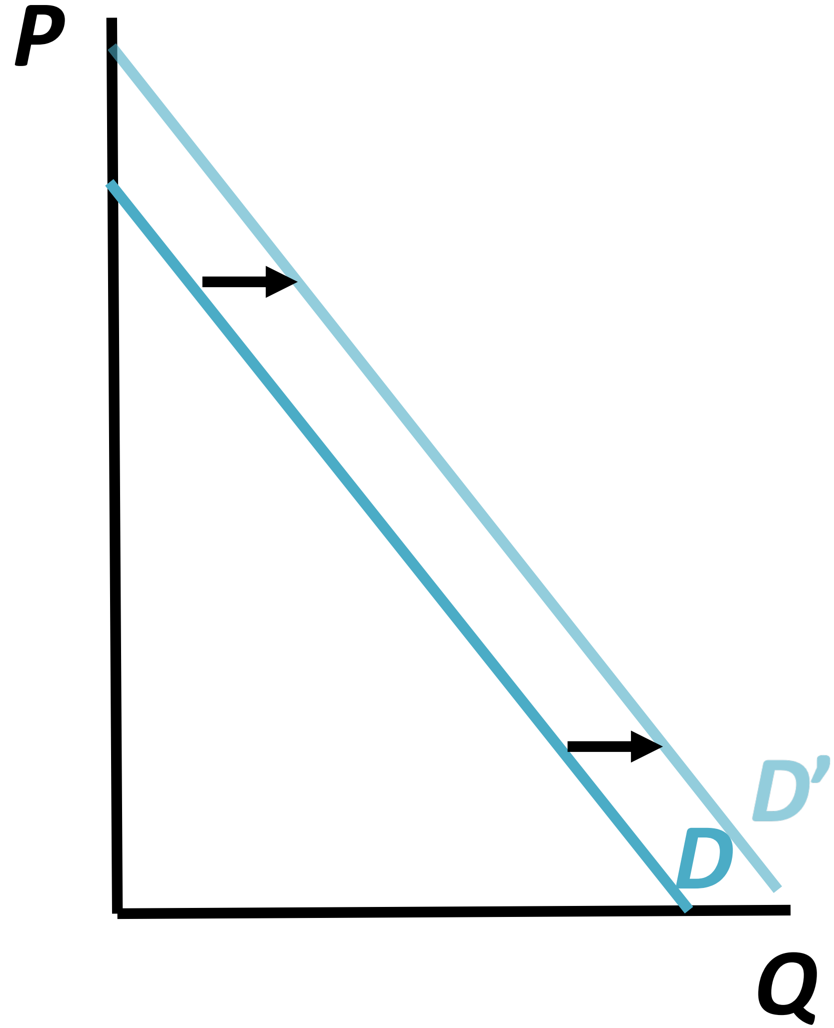
What happens to the demand for luxury goods when income increases?

Demand Shifts

Shock to income (Normal good):

$\uparrow M \Rightarrow \uparrow D_{LVbags}$

→*Demand for luxury goods shifts right (increases)*



Demand Shifts

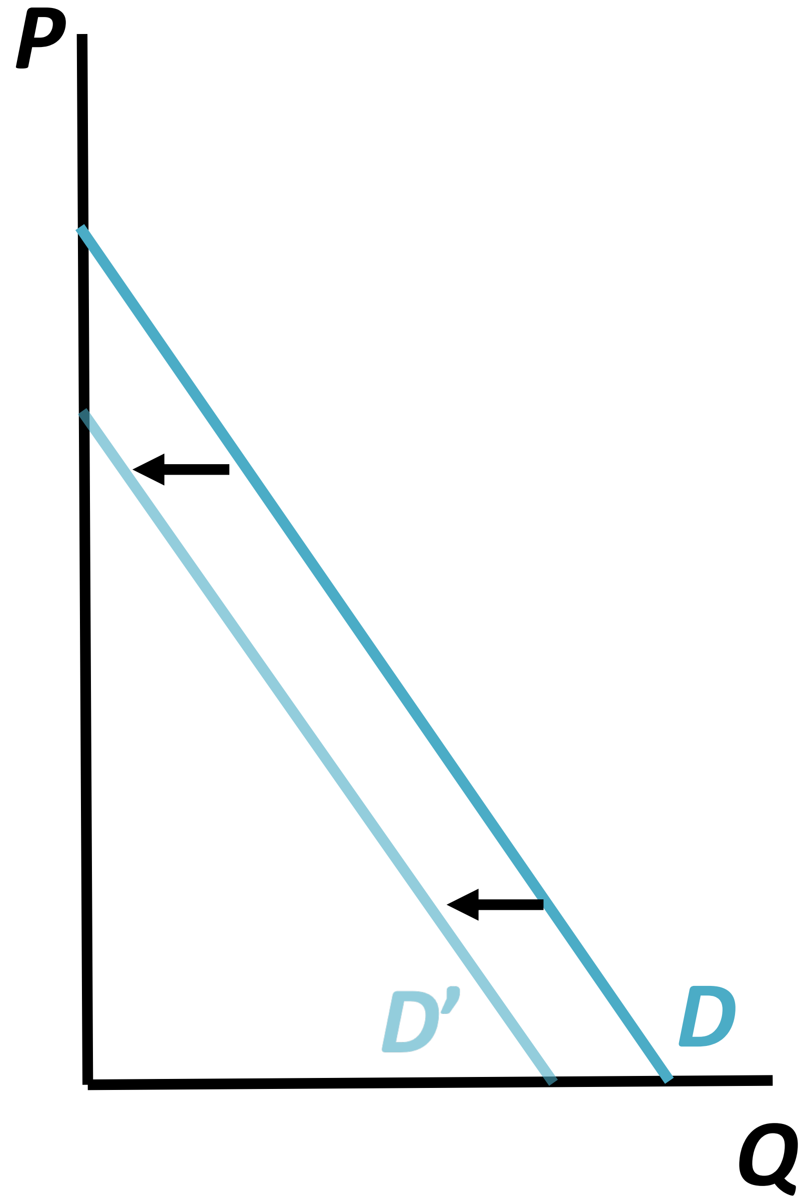
What happens to the demand for fake luxury goods when income increases?

Demand Shifts

Shock to income (inferior good):

$\uparrow M \Rightarrow \downarrow D_{fakeLVbags}$

Demand shifts left (decreases)



MINICASE: THE DEMAND FOR HEALTHY FOODS

Consumers Want Healthy Foods--And Will Pay More For Them

Nancy Gagliardi Contributor ⓘ
I explore food, culture and commerce.

Consumers' health is front and center for those who churn out the food products that will line supermarket shelves in the next year. At Campbell's, organic and all natural are the buzzwords as the company prepares to launch a variety of soups using USDA-certified organic ingredients and shelf-stable fruit and vegetable juices with no added sugar or artificial ingredients. At the 2015 Consumer Analyst Group of NY conference this week, many of the major food industry representatives echoed the health theme. Executives from Mondelez, for example, spoke about their recent acquisition of healthy lifestyle brand Enjoy Life Foods (which includes products for consumers who have allergies or want gluten-free and non-GMO fare), citing 30% growth of the US allergen-free segment. General Mills GIS +0% execs are betting on gluten-free, too: They cite the category as a bright spot with an estimated $8.8 billion in US retail sales in 2014 with predicted growth of $10.6 billion in 2015. The company is launching varieties of gluten-free Cheerios this July.

Global sales of healthy food products, in fact, are estimated to reach $1 trillion by 2017, according to Euromonitor. While the health fads and trends have come and gone (remember oat bran in the 1980s or low-fat everything in the 1990s?), this time the category appears to have serious stamina. Consider Nielsen's 2015 Global Health & Wellness Survey that polled over 30,000 individuals online and suggests consumer mindset about healthy foods has shifted and they are ready to pay more for products that claim to boost health and weight loss.

- Some 88% of those polled are willing to pay more for healthier foods.

- All demographics—from Generation Z to Baby Boomers--say they would pay more for healthy foods, including those that are GMO-free, have no artificial coloring/flavors and are deemed all natural.
- Functional foods—including foods high in fiber (36%), protein (32%), whole grains (30%) or fortified with calcium (30%), vitamins (30%) or minerals (29%)--that can either reduce disease and/or promote good health also are desirable.

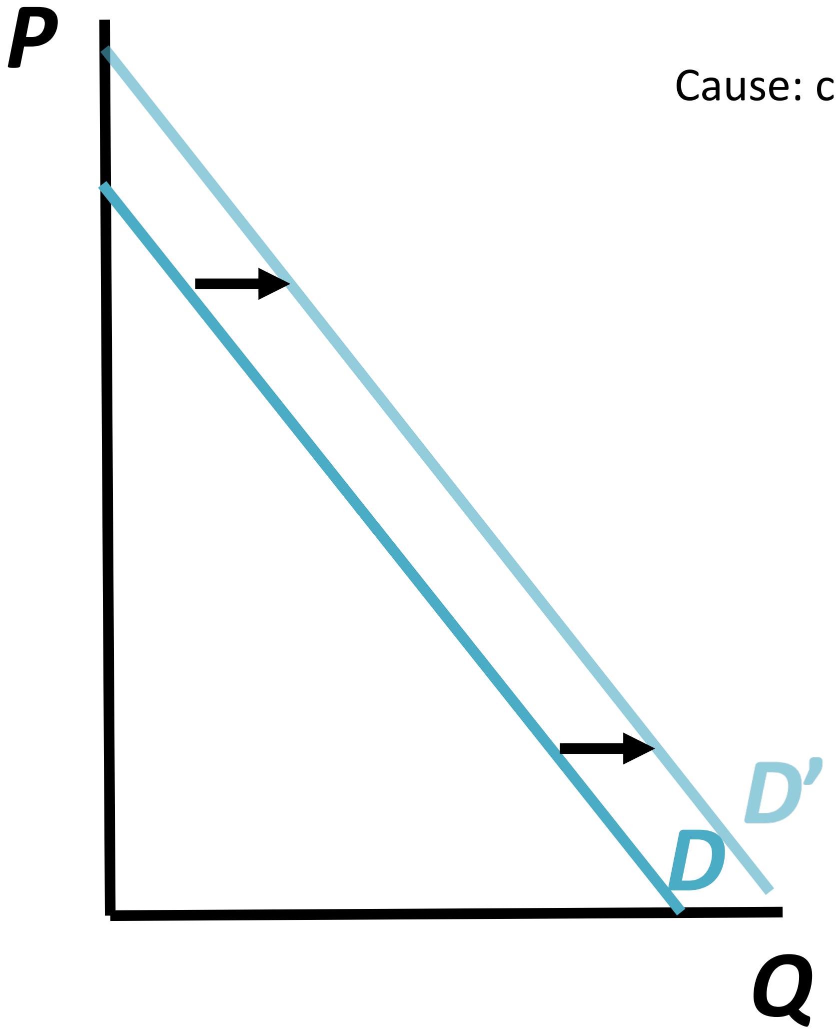
So what triggered the shift? James Russo, SVP, Global Consumer Insights at Nielsen offers perspective. "While economic concerns remain in the forefront for consumers, health and wellness concerns continue to increase in importance. The reasons vary from societal, demographic, technological, governmental and, most importantly, a shift in consumer focus on the role diet plays in health.

" "In fact, as consumers take more responsibility for their health food as medicine is becoming increasingly dynamic."

This idea of using food to manage health may, in part, help explain growing consumer interest in fresh, natural and organic products. Russo also suggests that consumers understand a food's nutritional value (in helping to lower blood pressure, for example), as well as overall health risk. Some of this response is partially rooted in growing corporate transparency about a product's health benefits. Yet Russo cautions that consumers are skeptical when it comes to food manufacturers' claims. "The claim has to be credible. A food high in sodium that promotes itself as rich in whole grains won't work. Consumers are very savvy."

While the food giants are taking big bets on health, the survey also points to something countless marketers already know: Consumers can be fickle. "There's always a bit of a disconnect with consumers. As it relates to diet, however, there is an aspirational component, and that's consistent. But there's a balance, too." In other words, they want healthy *and* indulgent products. So large fries and diet soda will probably still be on the menu.

The demand for healthy foods



Cause: change in preferences

Demand Functions

- A demand function is a mathematical representation of a product demand:

$$Quantity\ Demanded = D(\text{Price},\ \text{Other factors})$$

- From the demand function we can learn if products are substitutes or complements, and normal or inferior (look at signs)

Sample Demand Function

- Demand for pizza affected by: price of pizza, price of burgers, price of soft drinks (SD), consumer incomes (M or I)

Q^d_{pizza}=5-2P_{pizza}+4$P_{burgers}$-0.25P_{SD}+0.0003M

- Increases in the prices of pizza and soft drinks will decrease the amount of pizza that consumers demand
- Increases in the price of burgers or income will increase the amount of pizza that consumers demand
- When you draw it, fix the price of pizza, price of SD and income
  - E.g. If we fix M=$30,000; $P_{burgers}$=$0.5; P_{SD}=$4 then demand function will be:

$$Q^d_{pizza}=15-2P_{pizza}$$

Plotting a Demand Curve

- Assume linear function: $Q^d = a - bP$,

  where a, b are constant and positive terms.
- To plot a demand curve: use the **inverse demand function**: $P = a/b - Q/b$
- Y-intercept: a/b
- X-intercept: a
- Slope: $-1/b$

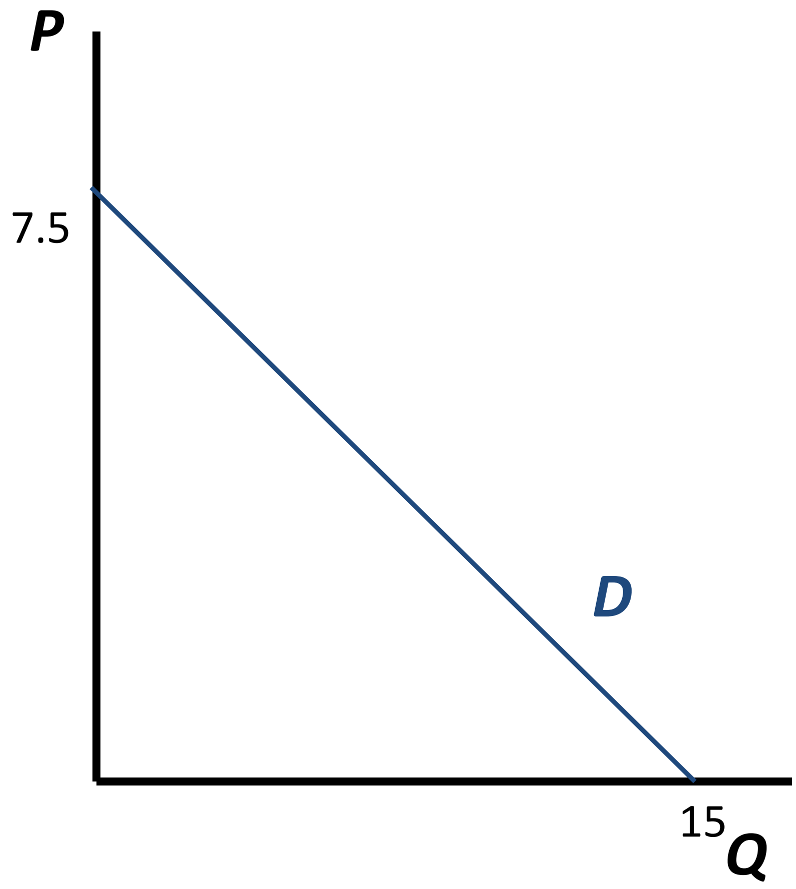
Demand Curve for pizza market

Q_{pizza}^{d}=15-2P_{pizza}

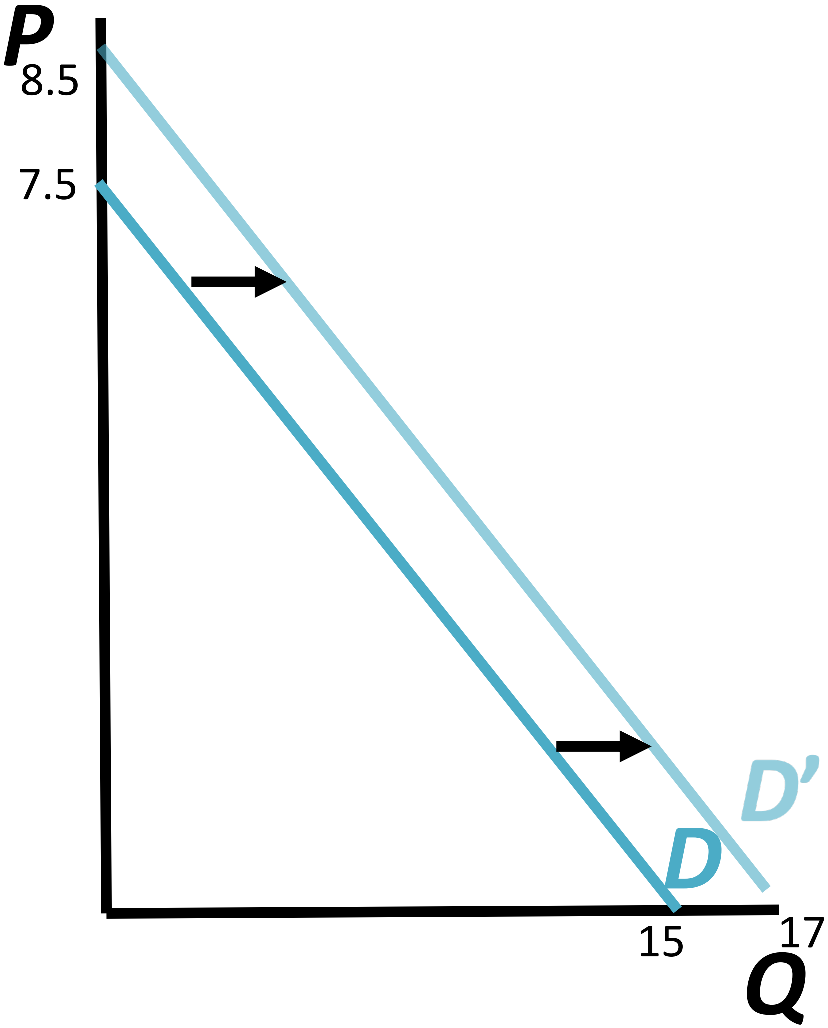
Inverse demand curve:
P_{pizza}=7.5-0.5 Q_{pizza}^{d}

Y-intercept: set Q_{pizza}=0
then P_{pizza}=7.5

X-intercept: set P_{pizza}=0
then Q_{pizza}=15



Shift in Demand Curve for pizza



If price of burgers increases from \$0.5 to \$1, then the new demand curve will be:

Q^{d}_{pizza}=17-2P_{pizza}

new Y-intercept:
P_{pizza}=8.5
new X-intercept:
Q_{pizza}=17

Supply Curve

Determinants of Market Supply

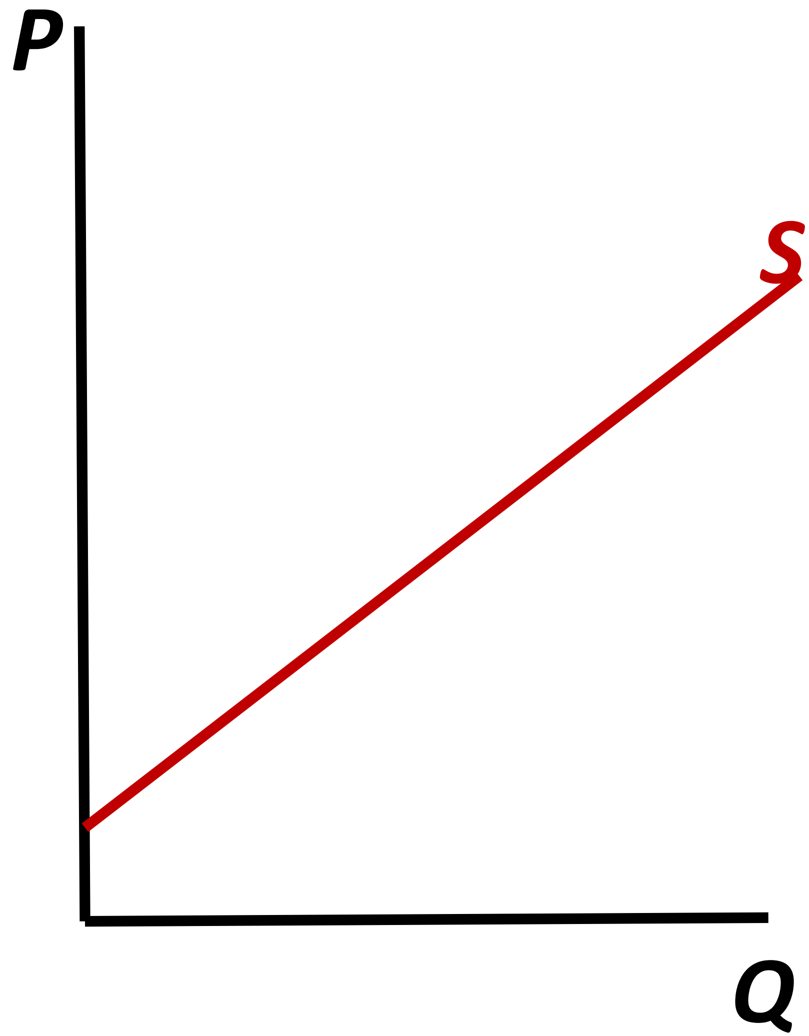
- Supply curve determined by:
  - Price of the good
  - Technology of the firms
  - Prices of inputs – e.g. labor
  - Government taxes or regulations
  - number firms

Supply Curve

- Product' s supply curve shows:

  How much sellers of the product want to sell at each possible price *holding fixed all other factors that affect supply*

- Upward sloping (selling the product is less attractive when the price is low than when the price is high)
- Where does it come from? Individual supply curve (cost minimization and profit maximization)



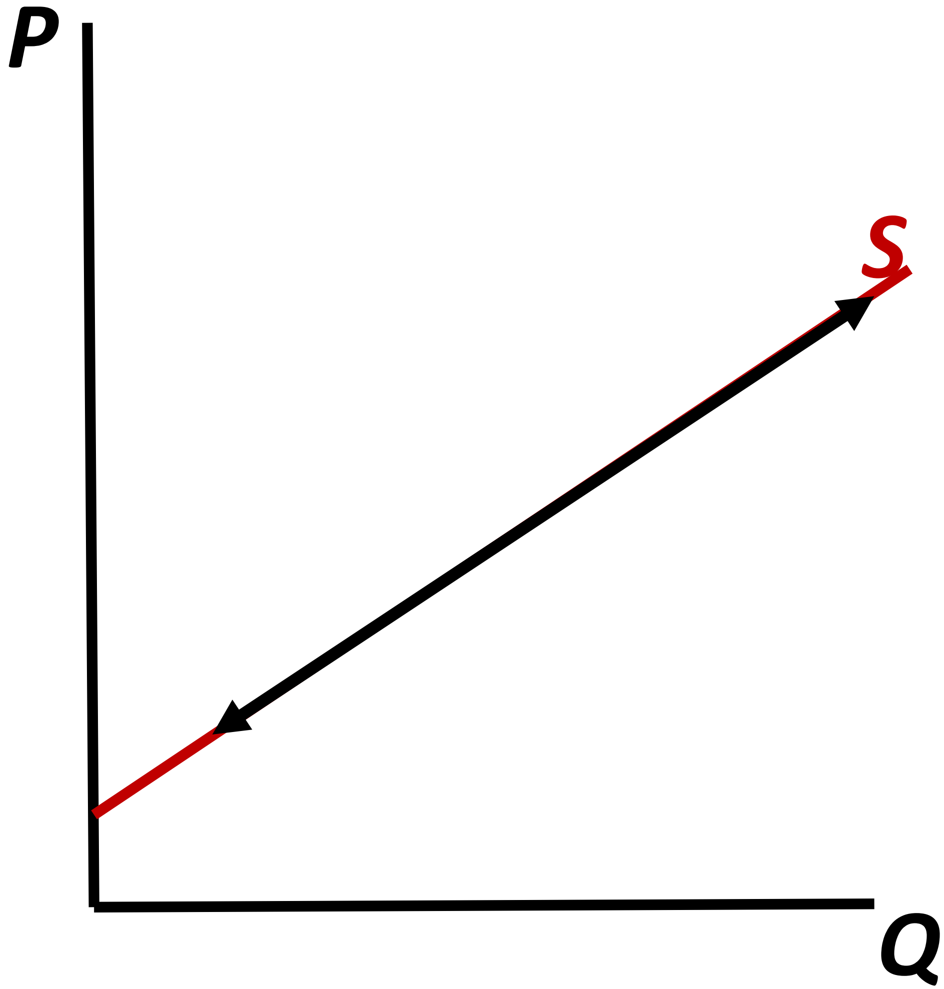
Shifts and Movements Along a Supply Curve

- Change in price of the product causes a ***movement along*** the supply curve
  - A change in the quantity supplied

- Change in another factor causes the entire supply curve to ***shift***
  - A change in supply

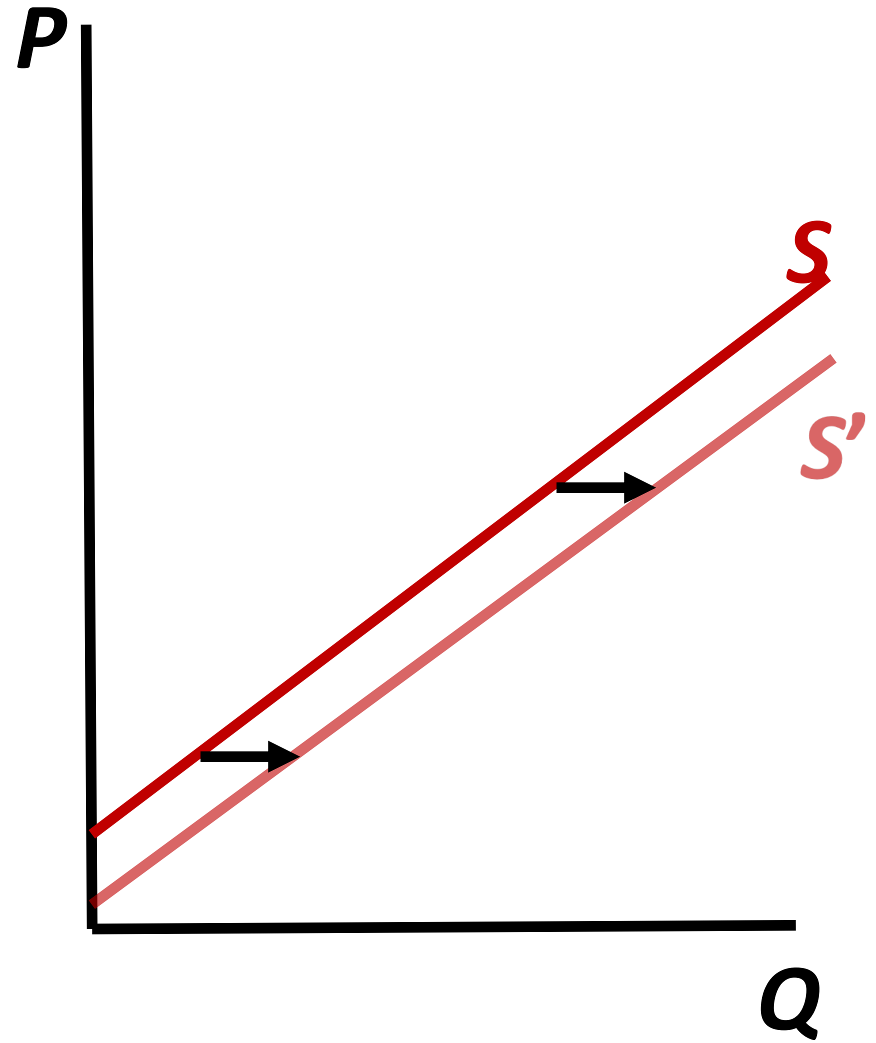
Movement along the Supply Curve

- Change in price of the product $\Rightarrow$ *movement along the curve*



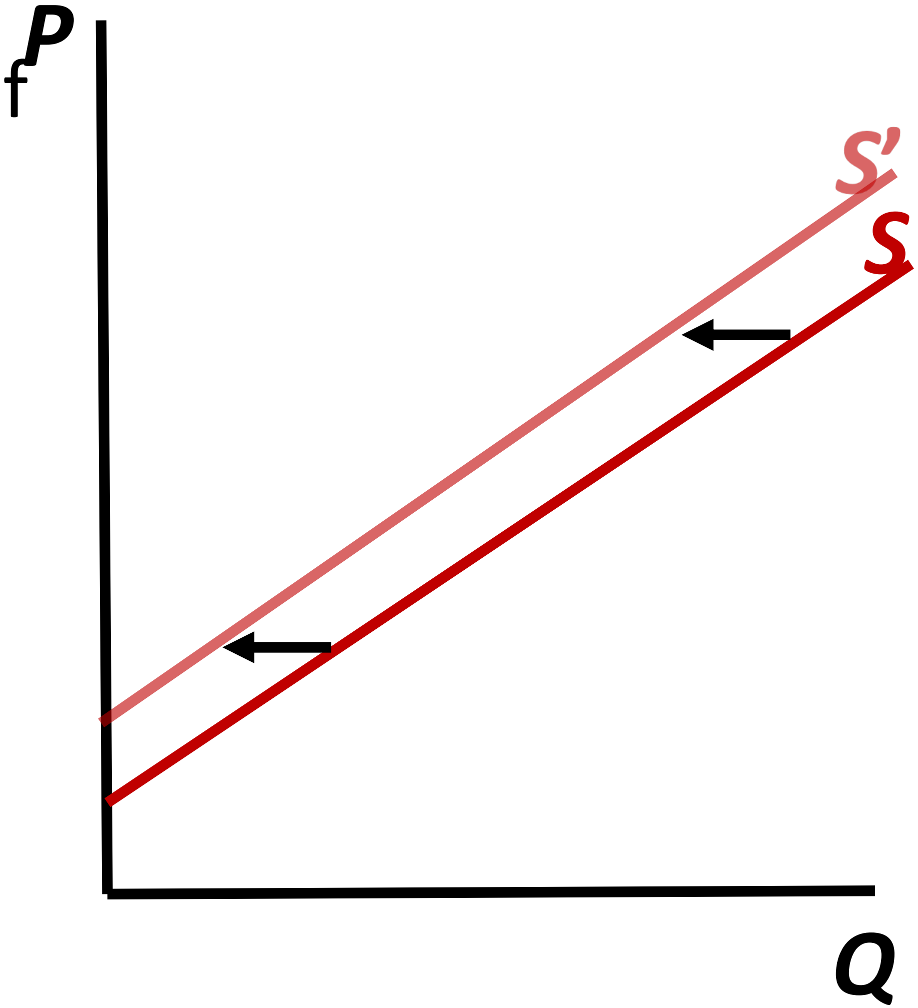
Supply Shifts: Increase

- Decrease in prices of inputs
- Decrease taxes/regulations
- Other factors



Supply Shifts: Decrease

- Increase in prices of inputs
- increase in taxes/regulations
- Other factors



Supply Functions

- Product's supply function is a mathematical representation of its supply:

$$Quantity\ Supplied = S(\text{Price},\ \text{Other factors})$$

Sample Supply Function

- Supply of pizza affected by: price of pizza, cost of tomato sauce, price of bread

$$Q^{s}_{pizza}=9+5P_{pizza}-2P_{tomato}-1.25P_{bread}$$

- Increases in the price of tomato sauce (input) and bread (substitute of production) will decrease the amount of pizza sellers supply
- Increases in the price of pizza will increase the amount of pizza sellers supply
- For example, if we fix P_{tomato}=\$2.5, P_{bread}=\$8, the supply function becomes:

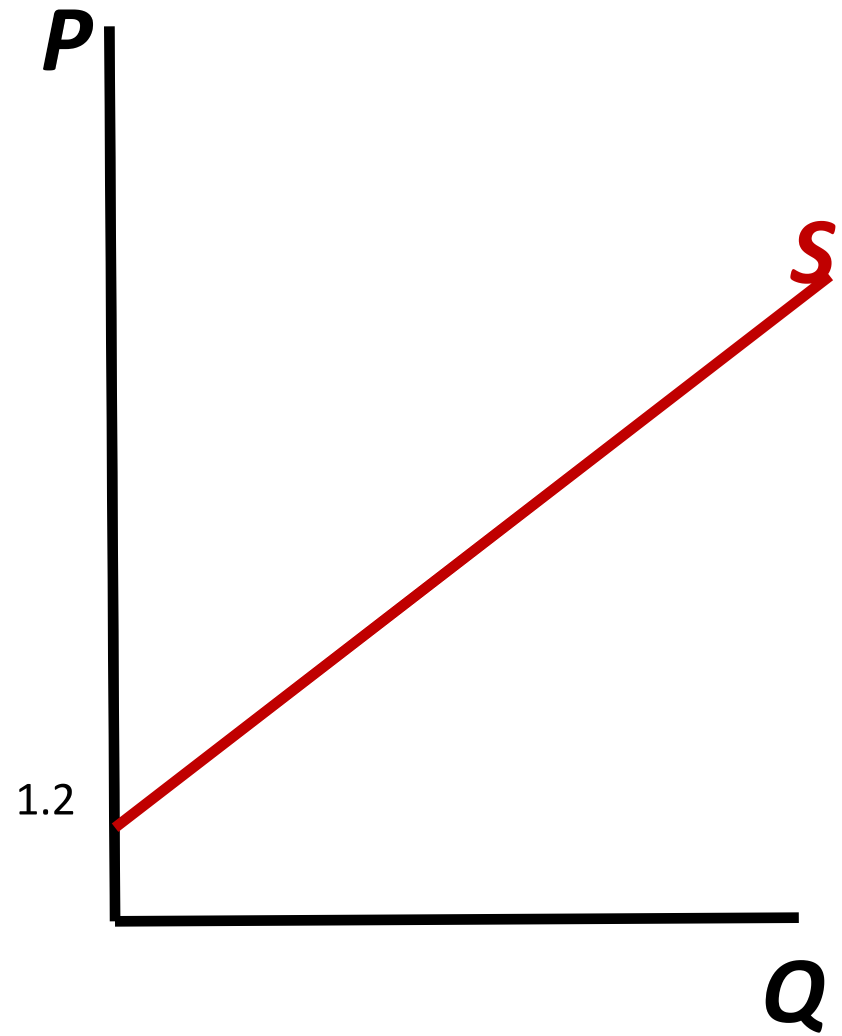
$$Q^{s}_{pizza}=5P_{pizza}-6$$

Plotting a Supply Curve

- Assume linear function: $Q^s = bP - a$ (where $b>0$)
- To plot a supply curve: use inverse supply function: $P = Q/b + A/b$
- Y-intercept: a/b
- Slope: $1/b$

Supply Curve for pizza Market

$Q^{s}_{pizza} = 5P_{pizza} - 6$

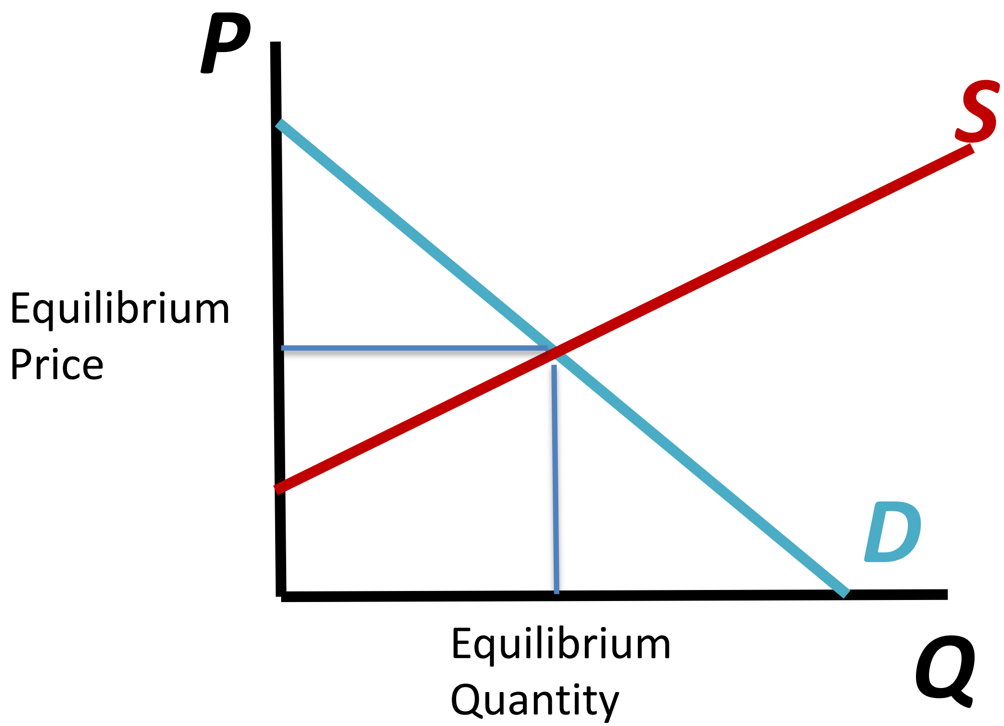


Y-intercept: set $Q_{pizza}=0$, then $P_{pizza}=1.2$

Slope: 1/5

Market Equilibrium

- The ***equilibrium price*** is the price at which the amounts supplied and demanded are equal
- Graphically, the price at which the supply and demand curves intersect:



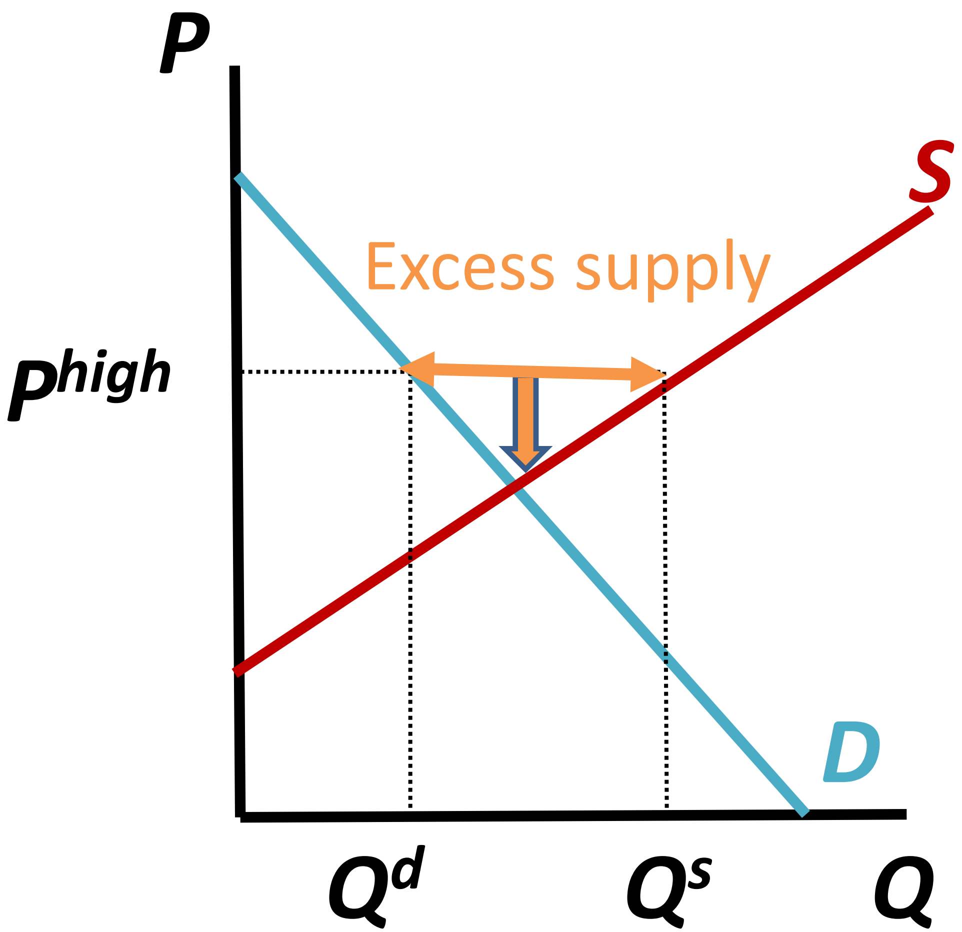
Market Equilibrium

Excess supply

$\Rightarrow$ sellers lower their prices

$\Rightarrow$ Q^s decreases and Q^d increases

$\Rightarrow$ lower excess supply, until it disappears



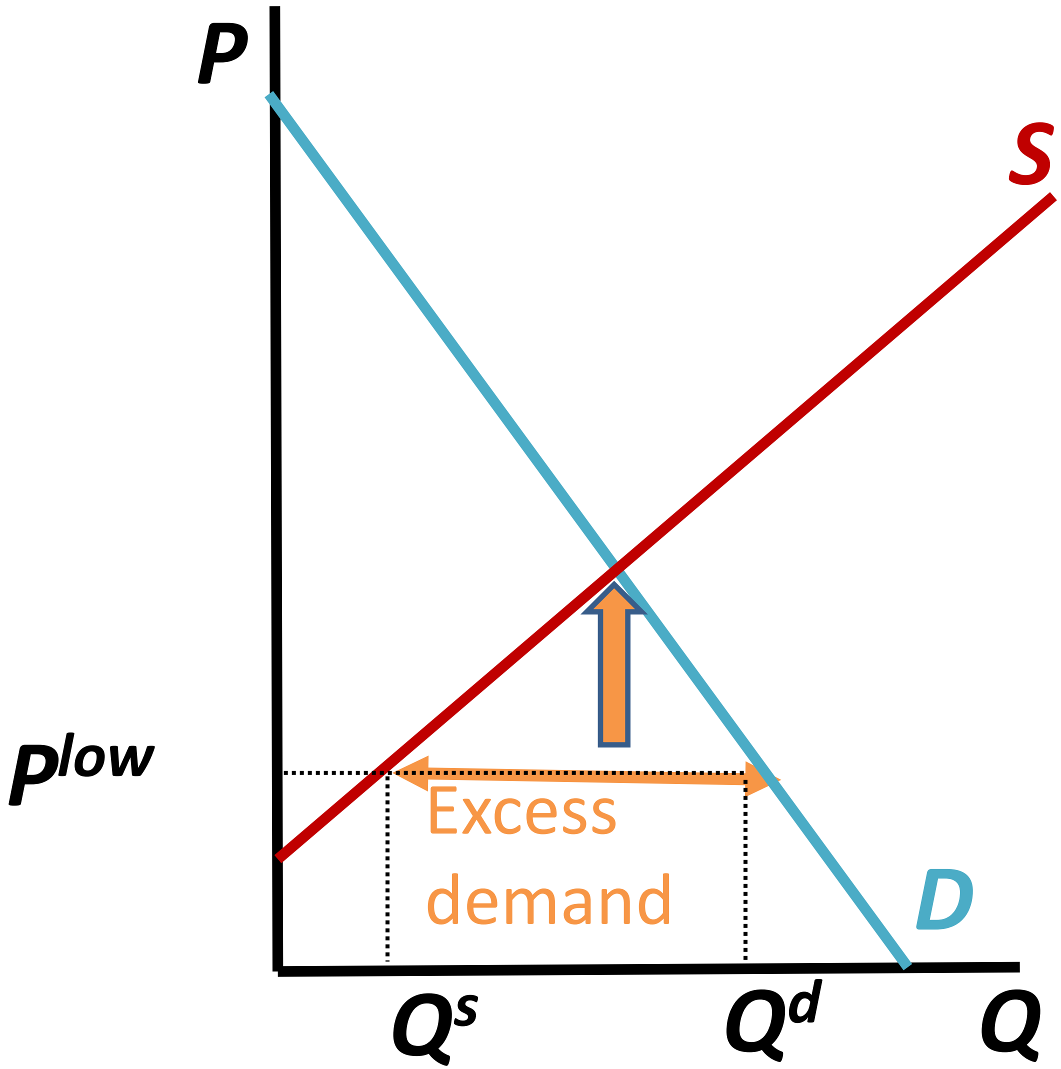
Market Equilibrium

Excess demand

$\Rightarrow$ buyers increase their bids

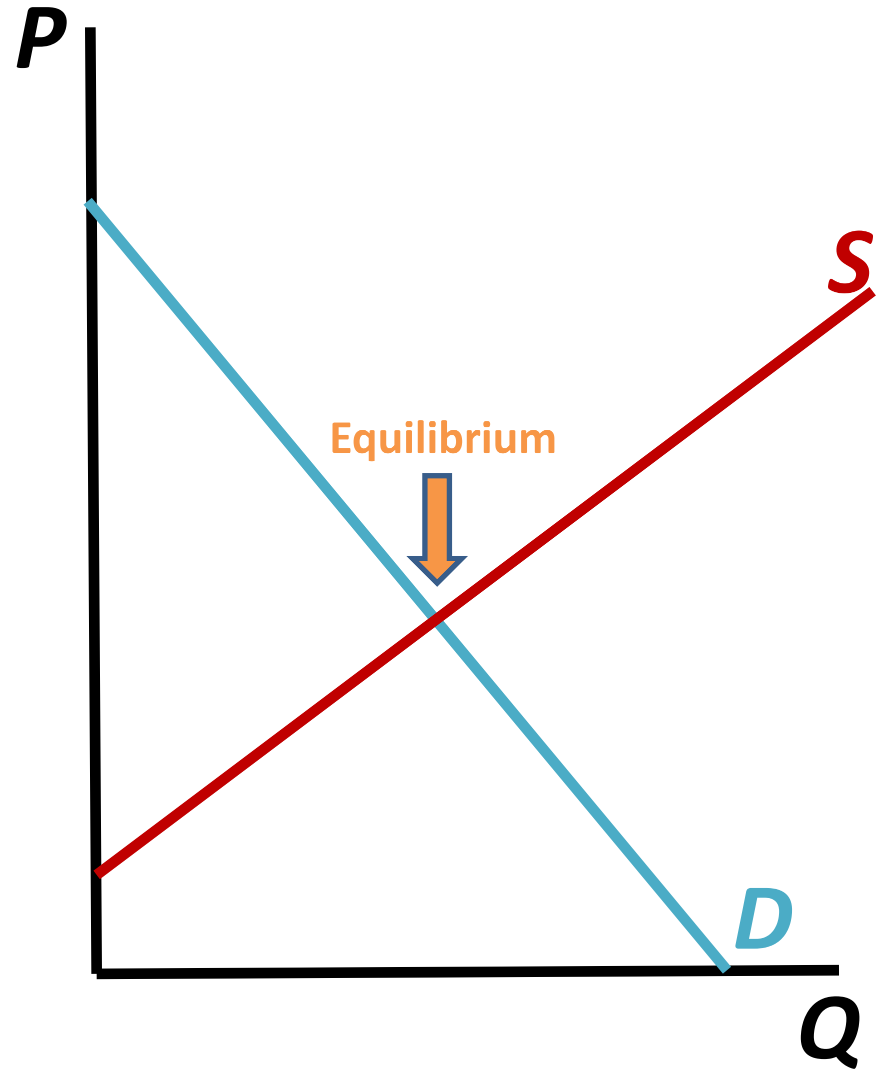
$\Rightarrow$ Q^s increases and Q^d decreases

$\Rightarrow$ lower excess demand, until it disappears



Market Equilibrium

$Q^d = Q^s$



Pizza market example

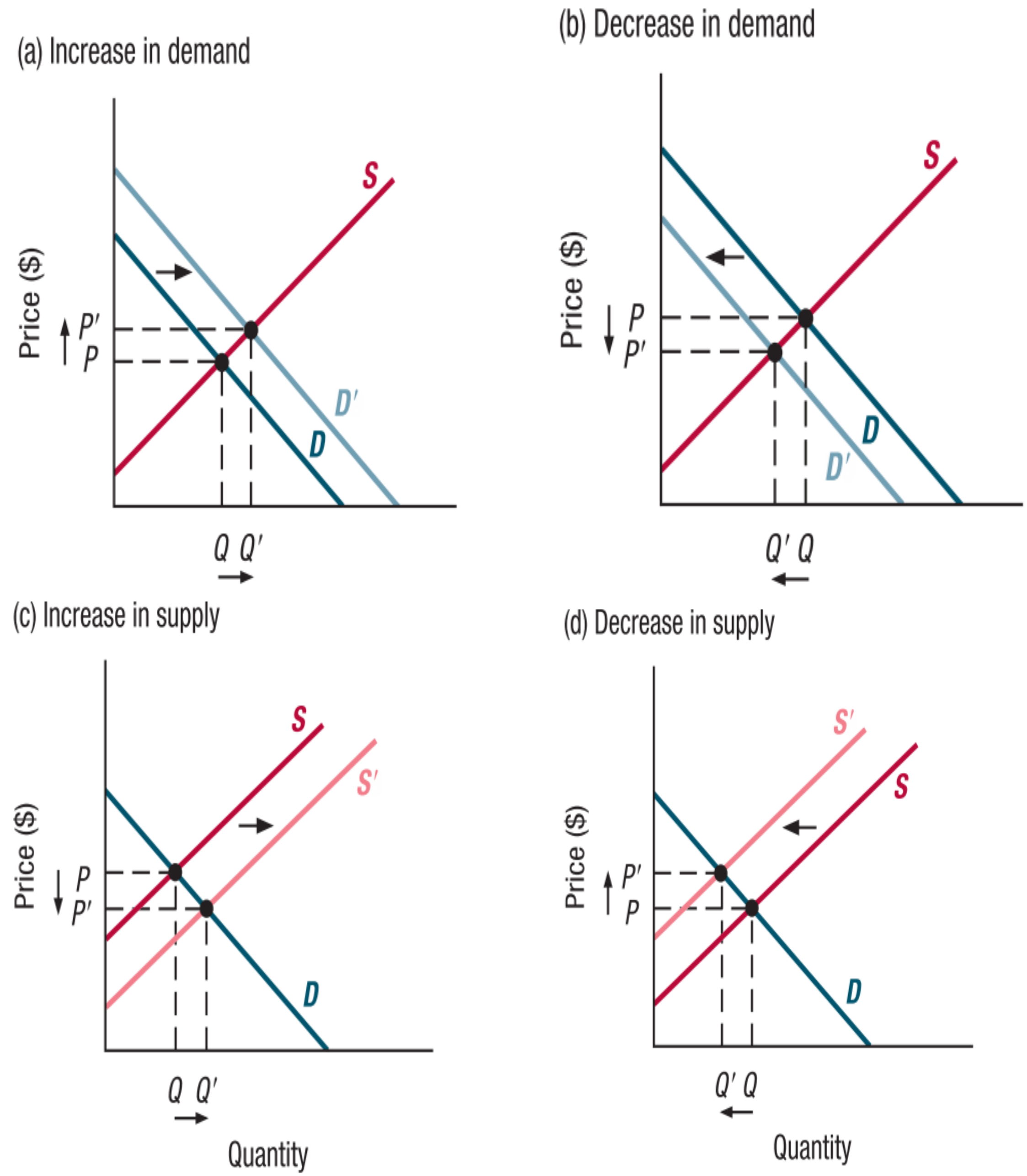
- Let's find the equilibrium price and quantity in the example:

$$Q^{d}_{pizza} = 15 - 2P_{pizza}$$

$$Q^{s}_{pizza} = 5P_{pizza} - 6$$

Changes in Market Equilibrium

- Changing market conditions alters the market equilibrium:
  - Changes in the determinants of supply (or demand) other than the product price cause the supply (or demand) curve to shift
- 4 cases: demand increases or decreases, supply increases or decreases


(a) Increase in demand
Price ($)
P′
P
S
D′
D
Q Q′
(b) Decrease in demand
Price ($)
P
P′
S
D
D′
Q′ Q
(c) Increase in supply
Price ($)
P
P′
S
S′
D
Q Q′
Quantity
(d) Decrease in supply
Price ($)
P′
P
S′
S
D
Q′ Q
Quantity

Changes in Market Equilibrium

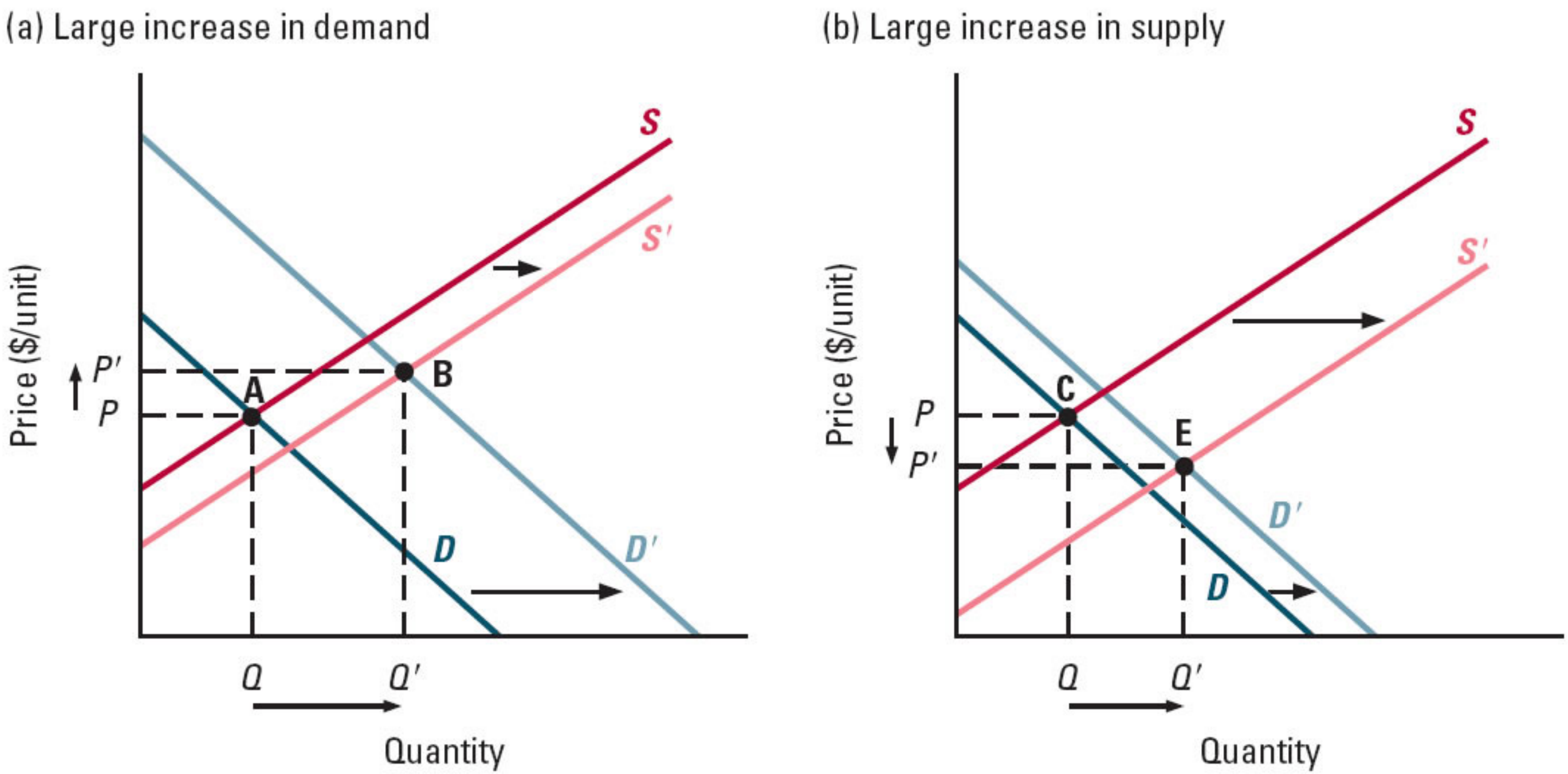
- Sometimes supply and demand will both shift
- Ultimate effect on equilibrium is combination of the separate effects of changes in demand and supply
- Will be able to determine the necessary direction of price or quantity movement, but not both

Increase in Both Demand and Supply

If supply and demand both increase:

- quantity will increase for sure
- what will happen to the price?
  - Demand increase pushes the price up
  - Supply increase pushed the price down
  - Net effect on price is ambiguous, depends on the relative size of the two effects on price

Increase in Both Demand and Supply



Effects of Simultaneous Changes in Demand and Supply

Source of Change	Effect on Price	Effect on Amount Bought/Sold
Demand and supply both increase	Ambiguous	Rises
Demand and supply both decrease	Ambiguous	Falls
Demand increases, Supply decreases	Rises	Ambiguous
Demand decreases, Supply increases	Falls	Ambiguous

Grain Market Minicase

FINANCIAL TIMES

Last updated: August 12, 2015 1:28 pm

Agricultural commodities feel the bite of weaker demand

Emiko Terazono

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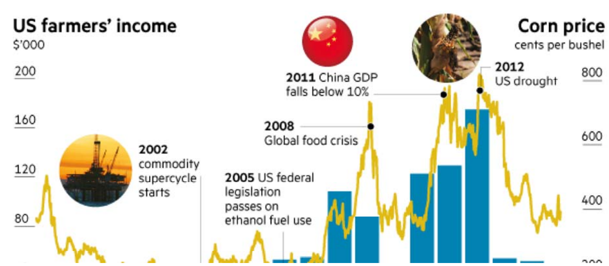
'Go-go years' end amid slower EM growth and higher output

As traders sound the death knell of the commodities supercycle, grains prices, which rallied on increased demand from emerging markets and periods of bad weather, also look to be heading for a period of weakness.

Along with oil and metals, agricultural commodities rode the great bull run in raw materials from the early 2000s. On top of new demand from developing countries, biofuel mandates also contributed to supply shortages.

However, as growth slows in China and other emerging economies, agricultural and rural economies are facing a "reset downward" says Professor David Kohl, a US agricultural economist formerly of Virginia Tech.

The "go-go years of 2007 to 2012 for the grain industry" were "set up by a convergence of timely events", he says.



The good years may have now come to an end. A slowdown in demand has coincided with increased production as farmers around the world have responded to higher prices, helped by ideal weather. Consecutive bumper crops over the past few years and the resulting high inventories around the world have depressed prices and incomes for farmers. The corn price, for example, is trading at $3.80 a bushel — less than half of the all-time record seen in 2012. In the US, one of the leading agricultural exporters, median net profits for corn farmers peaked in 2012 at $175,079 a household, falling sharply below $55,000 in 2013 and 2014, according to data from about 3,500 farmers compiled by the University of Minnesota.

As farmers' incomes have fallen, their spending has declined, squeezing profit margins in other industries that depend on it.

Grain market before 2015